

StateParl

Text corpus of speeches in German state parliaments

Documentation and Codebook

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Release Note

The release candidate builds upon the beta version of StateParl, but it incorporates significant improvements and refinements. These updates enhance data reliability, linkage with external data, and processing efficiency. The major changes in this release include:

- More data: StateParl now contains 14,092,664 cases (compared to previously 9,534,937 cases in the earlier beta version).
- Linkage with “StatePol” (Koch et al. 2024): In StateParl, persons are identified using *Länder*-specific patterns. In this version, we automatically matched these results with StatePol, a comprehensive dataset of German state politicians. Validation of the matching process (sample size: 3,330 distinct speakers) confirmed an accuracy rate of 99.34% correct matches, 0.24% unclear cases, and 0.45% false matches. Overall, 4,674 speakers could be identified and successfully linked with StatePol.
- Introduction of the protocols dataset: The metadata of the parliaments’ stenographic protocols are now stored in a dedicated dataset.
- Access to raw data files: StateParl now offers direct access to the originally scraped PDF files, providing full transparency and enabling further analysis. This also ensures the availability of and convenient access to the stenographic protocols, since some parliamentary web services only provide recent protocols online.
- Correction of issues related to “Zwischenfragen” (interposed questions) in Schleswig-Holstein: The beta version of StateParl encountered speaker misidentifications due to missing speaker names after “Zwischenfragen”. This issue has been addressed by storing paragraph-based position metadata, allowing for the correct identification of indents and preventing misattributions.

Further medium and smaller improvements are presented in the documentation.

Abstract: StateParl comprises the parliamentary speeches of MPs and government representatives in all 16 German state parliaments from 1.1.2000 to 31.12.2023. The database consists of 14,092,664 paragraphs of parliamentary speeches. After introducing the general idea behind StateParl and its advantages (1.), this documentation presents the final data structure of StateParl (2.) and explains the construction of the database in detail: document scraping (3.1), document preparation (3.2), processing (3.3), data validation (3.4), and current limitations (3.5). The documentation then proceeds with the changelog, in which we outline the improvements compared to the earlier beta version (4.). The documentation also contains the codebook (5.).

Please note that the data are not only accessible through this database but also via an Application Programming Interface (API) which offers easy access to the up-to-date version of the data. A search interface based on this API can be accessed via <https://stateparl.de>.

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1. Introduction

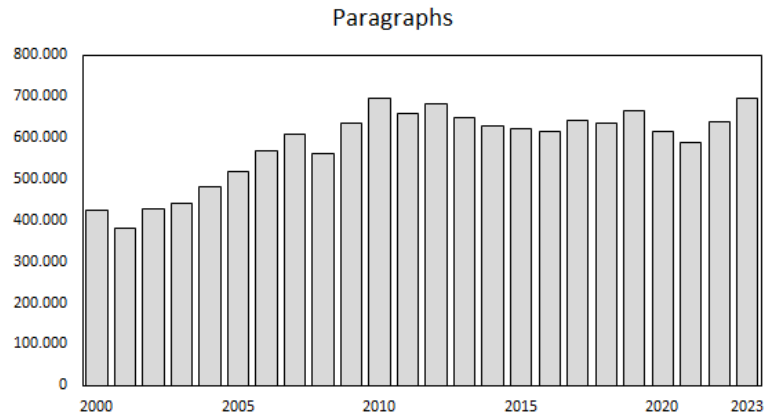
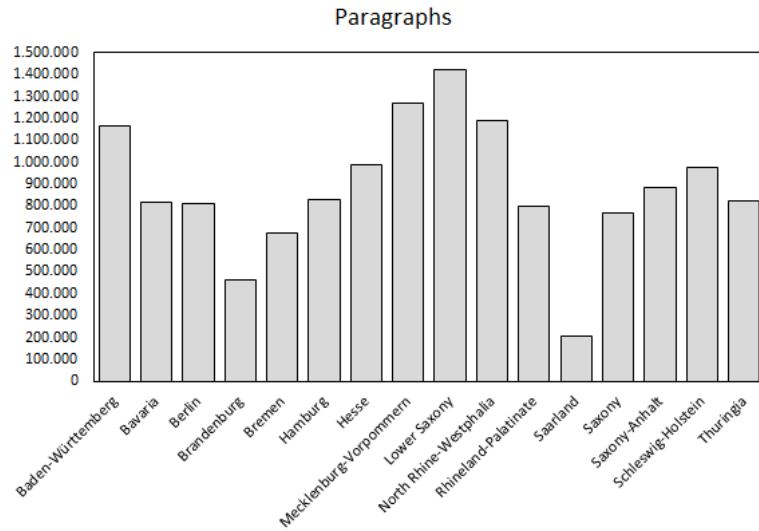
StateParl comprises the parliamentary speeches of MPs and government representatives in all 16 German state parliaments from 1.1.2000 to 31.12.2023. These speeches are recorded in stenographic protocols, with one protocol corresponding to each plenary session in the state parliaments.

Stenographic protocols have become an important source for researchers, students, and journalists. StateParl integrates these protocols in an accessible, coherent, and machine-readable format that enables systematic applications.

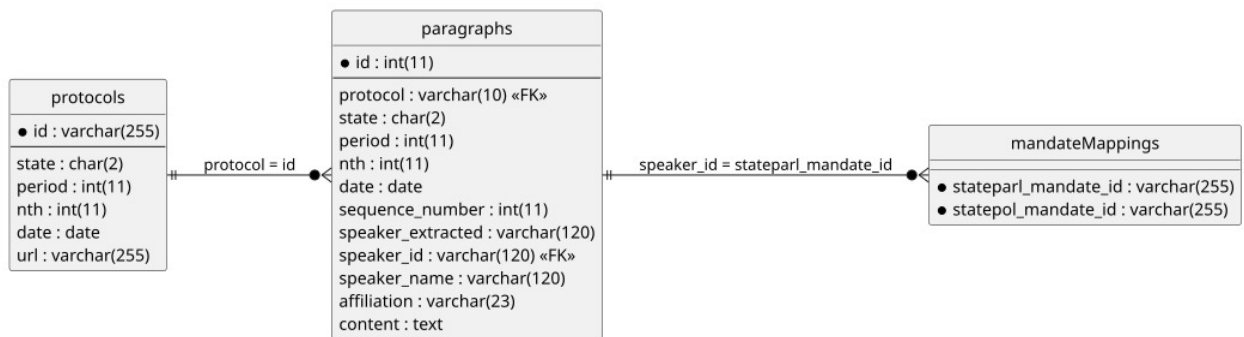
The stenographic protocols include speech acts directed at the public in which political actors explain their positions on various policies, measures, and programs. They refer not only to decisions and actions of the respective state government, but also to the political actors and actions in other states (*Länder*), at the federal level, or at the European Union level. Therefore, parliamentary debates and speeches in state parliaments are a valuable source to explore the logics of the multi-level institutional setting. However, they not only facilitate investigating parliamentary discourses and communicative practices. Reflecting the motivations and interests behind decisions made by governments, parliamentary speech acts allow for investigating policy-making processes. Since in Germany the parliamentary party groups (PPGs) determine the MP who takes the floor in the debate, parliamentary debates and speeches can also be leveraged to identify party positions, party competition, and intra-party conflicts.

StateParl offers several significant advantages for anyone interested in researching these topics:

- **Transparency and quality:** Data were validated to ensure the quality of the database. Adhering to Open Science standards, there is a transparent explanation of the methodological approach, so researchers, students, and anyone interested can understand how the data were generated. This approach facilitates data reuse.
- **Comprehensive coverage:** StateParl encompasses the parliamentary debates in all 16 state parliaments for a long period. It also uses paragraphs as the unit of analysis, following the structure of the original stenographic protocols. Altogether, StateParl thus allows for fine-grained, and wide-ranging analyses across the entire German federal system (see Figures 1 and 2).
- **Versatility for research approaches:** The structure and content of StateParl support a wide array of research methodologies. It is equally suitable for qualitative analyses, such as discourse analysis or case studies of specific debates or policy areas, and quantitative approaches, including statistical analyses of speaking patterns, topic modeling, or large-scale text analysis. StateParl also facilitates comparative studies and longitudinal analyses. This versatility makes StateParl a valuable resource for researchers across multiple disciplines, such as political science, communication science, sociology, linguistics, legal studies, data science, and economy.

Figure 1: Number of paragraphs over time**Figure 2: Number of paragraphs across parliaments**

2. Final data structure



3.1. Document scraping

We accessed the stenographic protocols from the 16 web services of the state parliaments. Initially, we scraped the metadata of the protocols, using various techniques (Table 1). The relevant techniques vary due to differences in how each website presents data.

Table 1: Techniques to retrieve the stenographic protocols

Parliament	Technique
Baden-Württemberg	HTML Scraping
Bavaria	HTML Scraping
Berlin	XML Overview Files
Brandenburg	XML Overview Files
Bremen	API Get Request
Hamburg	API POST Request
Hesse	HTML Scraping
Mecklenburg-Vorpommern	HTML Scraping
Lower Saxony	HTML Scraping
North Rhine-Westphalia	HTML Scraping
Rhineland-Palatinate	HTML Scraping
Saarland	API POST Request
Saxony	HTML Scraping
Saxony-Anhalt	HTML Scraping
Schleswig-Holstein	API POST Request
Thuringia	Headless Browsing

Note: For Saxony and Thuringia, we enrich the scraped lists of protocols with protocols that we have previously collected and are not available anymore at the web services of the two parliaments.

We created an extra protocols dataset (see final data structure, 2.), containing information on protocols' metadata. Each document is represented by a row in the dataset. Once all the information on the available documents had been stored, the files were downloaded. This, as well as any other use case where data or header information were requested, was done by using the python *requests* library. Optical Character Recognition (OCR) was used to add a text layer if necessary due to the absence of machine-readable text or the detection of charset errors. These were detected by catching encoding errors.

```
def check_charset(self, text):
    if isinstance(text, list):
        text = ''.join(text)
    try:
        text.encode('utf-8')
    except UnicodeEncodeError:
        return False
    return True
```

OCR was done using the *OCRMyPdf* library², which has returned the best results while testing free OCR libraries. *OCRMyPdf* uses the *Tesseract* engine³, and it produces a searchable PDF/A document. The text was added as an additional layer below the images.

```
ocr = ocrmypdf.ocr(
    file_path,
    file_path,
    deskew= True,
    clean= True,
    language= 'deu',
    optimize= 1,
    force_ocr= True,
    jobs=10,
)
```

To continue with text preparation, we needed to extract the text. We used the *PyMuPDF*⁴ library because it has returned the best results for multi-column documents. The text on each page was extracted and collected in a list that we later merged into a text block. If there were still charset errors, we tried to fix them with *Ghostscript*⁵.

```
def pdf_to_text(self, path, pages = False):
    pages = int(pages)
    paragraphs = []
    with fitz.open(path) as doc:
        for i, page in enumerate(doc):
            if pages and i >= pages:
                break

            raw = page.get_text("blocks")
            for p in raw:
                paragraphs.append(p[4])

    return paragraphs
```

² <https://pypi.org/project/ocrmypdf/> [15.07.2024].

³ <https://github.com/tesseract-ocr/tesseract> [15.07.2024].

⁴ <https://pypi.org/project/fitz/> [15.07.2024].

⁵ <https://pypi.org/project/ghostscript/> [15.07.2024].

Each paragraph was handled separately and comes with metadata on the respective position to handle speech interruptions marked by indent. The prepared protocol files served as foundation for the next stages of database construction.

3.2. Preparation of documents

The documents were prepared for later storage in a database. The goal was to make the texts as “clean” as possible and to extract metadata. There are two types of metadata to identify: *document-wide metadata* such as the legislative period or the date of the plenary session the protocol refers to and *paragraph-specific metadata* such as the speaker.

Firstly, we cleaned the text on a protocol level. This was done by running a non-specific function that performed several tasks. It removed unnecessary line breaks and fixed character set errors. Tables of contents were removed as far as they could be identified. We also identified “interjections” of the speech, under which we summarize any interruptions of the speech, such as interjections (“*Zwischenrufe*”) by other MPs, applause, and explanations due to protocol purposes (such as “ringing the bell” by the president and the like). This was important to avoid assigning, for example, an interjection as part of the current speech or wrongly defining it as a new speech.

Secondly, we ran *Länder*-specific functions to remove text such as page headers and footers, and other paragraphs without language content. Therefore, text elements that are not part of the speeches were removed as far as possible. This also included all appendices of the protocols (in which, for example, written questions of the MPs or written answers of the government are included). The updated text was saved in a separate *JSON*⁶ file. Preparation resulted in a “clean” text file with separated paragraphs of the speeches, while paragraphs that are not part of the speech, like interjections, were marked.

3.3. Processing

The documents were then processed. This step included the identification of when the speaker changes in the protocol, the collection of metadata, as well as data storing. We used a *MariaDB* setup for storing the data. *MariaDB* is an open-source database which helps handling the big amount of data by offering SQL-based request methods and indices. As a relational database, it is also convenient for storing row-based data. Data were processed for each Land individually because of different patterns in the protocols.

Following the original format of the stenographic protocols, StateParl treats the paragraph as unit of analysis. Processing was thus done at the paragraph level, and it consisted of three different steps: In the first step, the paragraphs were cleaned by removing unnecessary white space and line breaks within words caused by the PDF representation of the data. In the second step, speakers and affiliations were identified. In the current version of StateParl, only the paragraphs of parliamentary speech are assigned to speakers. Interjections are not yet linked to speakers (see limitations, 3.5). Persons were identified using *Länder*-specific patterns. The results were then matched with information in StatePol,⁷ a comprehensive dataset

⁶ JSON (JavaScript Object Notation) is a lightweight data format that stores information in a way which is easy for both humans to read and computers to process. It's commonly used to transmit data between a server and a web application.

⁷ Koch, E., Kuhlen, D., Müller, J. *et al.* (2024): StatePol – Eine Datenbank zu den Mitgliedern von Regierungen und Parlamenten in den 16 Bundesländern. *Politische Vierteljahresschrift*. <https://doi.org/10.1007/s11615-024-00528-z>.

of German state politicians. The merge with StatePol was done automatically. The results were validated (99.34% correct, 0.24% unclear, 0.45% false) with a sample size of 3,330 distinct speakers. 4,674 speakers were identified. StateParl consists of 9,210 distinct speakers. The ones not covered are either out of the scope of StatePol or could not be identified automatically.

If the application could not identify a new speaker and the paragraph was not marked as non-speech (e.g., an interjection), but subsequent paragraphs followed, it is assumed that the next paragraph continued the current speech and is therefore attributed to the same speaker.

Affiliation describes the role in which the person is speaking in parliament. The application distinguishes between the patterns of speaker affiliations that are presented in Table 2.

Table 2: Patterns of speaker affiliations

Pattern	Explanation
Default	Members of the parliament in their role as members of their PPG or as independent MPs (<i>“fraktionslos”</i>)
Presidium	- President or Vice-President of the parliament acting as chair of the plenary session and not as member of their PPG. - Clerks: In some cases, clerks (<i>“Schriftführer”</i>) spoke in their role as members of the chair. They are thus assigned to the presidium affiliation.
Government	- Heads of government: Minister Presidents, Governing Mayors, First Mayors, Presidents of the Senate, and Mayors - Cabinet members: Ministers and senators as well as state secretaries
Others	- Ombudsmen and ombudswomen, data protection officers, presidents of Constitutional Courts, presidents of Court of Auditors, other guests speaking in the parliament such as representatives of NGOs - Rapporteurs: Sometimes there was no indication of a PPG, but the speaker's role as rapporteur was mentioned. In these cases, we assigned them to the affiliation "others" because, based on the protocol, the PPG is unknown. When both rapporteur status and affiliation were indicated, the default pattern was used because the speakers were speaking in their role as a member of a PPG.

In the third step, we normalized affiliations to ensure consistency in the database. The labels of these affiliations were vastly different across parliaments. This applies also to the names of the PPGs. For the FDP, for example, we find several labels, such as *“Freie Demokratische Partei”*, *“FDP/DVP”*, *“Freie Demokraten”*, or *“F.D.P.”*. Details regarding the affiliations of the speakers can be found in the codebook (5.).

At this point, all necessary metadata were available. From the document context, we knew the date (*date*), the session number (*nth*), the legislative period (*period*), and the parliament (*state*). For each paragraph, we knew the content of the speech (*content*), and for all paragraphs of parliamentary speech also the speakers (*speaker_extracted*) and their affiliation (*affiliation*). For all paragraphs, we knew their incremental numbers in the context of the document (*sequence_number*). This information is inserted into a database. MariaDB added a unique auto-incrementing id (*id*) (Table 3).

Table 3: MariaDB table

Field	Type	Null	Default
id	int(11)	NO	NULL
protocol	varchar(10)	NO	NULL
state	char(2)	YES	NULL
period	int(11)	YES	NULL
nth	int(11)	YES	NULL
date	date	YES	NULL
sequence_number	int(11)	YES	NULL
speaker_extracted	varchar(120)	YES	NULL
speaker_id	varchar(120)	YES	NULL
speaker_name	varchar(120)	YES	NULL
affiliation	varchar(23)	YES	NULL
content	text	YES	NULL

Sometime, the PDF files of the stenographic protocols are not accessible via the web services of the state parliaments (e.g., because they only provide the ones of the most recent legislative periods, or they change the software). To secure the availability of all original protocols covered by StateParl, we provide a caching server from which users can access the PDF files. To this end, users can insert the protocol ID into an URL in the following pattern: <https://assets.stateparl.de/protocols/{protocol ID}.pdf>

For the stenographic protocol that we use in the codebook (5.) – 84th session of the 18th legislative period of the state parliament in Berlin, the URL is: https://assets.stateparl.de/protocols/be_18_84.pdf

3.4. Data validation

A full evaluation was not possible due to the size of StateParl. Therefore, we drew samples. To ensure that possible error patterns were mapped across all legislative periods and state parliaments, we implemented a two-stage clustered randomized stratified selection. In the first stage, a proportional weighting was carried out according to the size of the corpus for each state parliament. In the second stage, the number of paragraphs within the respective corpus was weighted according to the legislative period. This method helps maintaining the relative importance of each state parliament and period in the analysis.

$$w_{s,p} = \frac{n_{s,p}}{T}$$

Representation of sample weighting. We divide the number of records (n) per state parliament (s) for a given period (p) by the total number of records (T). The result is a weight (w) for each state parliament (s) and period (p) combination.

In a first validation, a total of 28,543 samples (at that time, this represented around 0.3% of the total number of paragraphs) were drawn. The validation took place from April to June 2024 and was carried out by research assistants at the Otto Suhr Institute of Freie Universität Berlin. For the validation, an online application was developed (Figure 4). This application also reported the errors for each person involved in the validation process as well as for each parliament. To ensure standardized validation among the

team members, guidelines and introductory training were provided. Interim evaluations also helped to maintain the quality of the validation. 96.3% of the samples were evaluated as correct, while 2.49% of the samples were evaluated as false. 1.21% of the samples referred to other parts of protocol like table of contents and annexes, which were still in the database despite protocol cleaning. There were mainly speaker and affiliation errors. A few errors regarding the state parliament, legislative period, number of the protocol, and date were identified as well. Prior to the release of the beta version, we could already fix several of these errors.

Figure 4: Web application for data validation

Explanation: On the left side, the buttons can be moved to false (*“falsch”*) if metadata were wrongly retrieved from the protocol. This allows more specific validation, since we also know *what* was wrong in a particular case. Cases could also be sent to the team for discussion (*“An Teamleitung schicken”*), be commented on (*“Kommentieren”*), and be marked as “trash” (*“Ist Trashprobe”*) if they referred to other parts of protocol than debates (for example to the protocol’s table of contents or annexes). On the right side, the application automatically opens the protocol at the correct position, making the validation more precise, easier, and less time-consuming.

In a second validation, using the same method for sampling, we drew 14,687 samples (equivalent to around 0.1% of the total number of paragraphs). The validation took place from December 2024 to January 2025 and was again carried out at the Otto Suhr Institute of Freie Universität Berlin. The second validation delivered even better results, as 97.61% of the of the samples were evaluated as correct, 1.23% as false, and 1.15% of the samples referred to other parts of protocol than debates and speeches.

3.5. Current limitations

While StateParl provides a comprehensive collection of parliamentary debates and speeches in the 16 German state parliaments, it is important to acknowledge certain limitations:

- Document availability: Within the period covered by StateParl, some stenographic documents are missing from the database due to their unavailability in the web services of the state parliaments. This results in some gaps in the coverage. We have already identified the affected state parliaments and periods (Figure 5). This lack of coverage may be corrected by contacting these parliaments directly. A promising initiative is the “Parlamentsspiegel” (<https://www.parlamentsspiegel.de>) that collects and provides all PDF documents of the German state parliaments since December 1978.
- Interjections are not assigned to specific speakers. For those who want to research these forms of parliamentary behavior, for example to study patterns of who is interrupted and by whom, information about the speakers is needed. Moreover, these non-speech paragraphs currently comprise different forms of parliamentary behavior, including interjections, applause or other actions such as bell-ringing by the president. These varieties of behavior must be further distinguished for meaningful analyses. Due to the variety of notations, this cannot be captured by the speaker-affiliation syntax used for the speech-based paragraphs.

Figure 5: Coverage of parliaments

	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
bb	86	83	91	23	54															
be												32	80	84	84	82	48			
bw										105	109	112	150	146	106					
by												115	123	124	132	147	26			
hb													78	83	86	81	82	46	15	
he															150	19	150	150	146	23
hh														100	56	99	72	100	100	91
mv	85	82	127	126	126	83														
ni												129	137	153	140	144	49			
nw										145	150	149	57	143	170	83				
rp											128	111	105	110	117	73				
sh												106	135	123	79	145	148	71		
sl											1	33	58	64	31					
sn		138	99	90	91	1														
st	73	74	89	107	126	76														
th	99	112	161	162	136	1														

■ Available protocols
 ■ Missing protocols
 ■ Untargeted / future periods

Note: Numbers and information refer to the legislative periods of the respective parliaments.

4. Changelog

This release candidate of StateParl was comprehensively improved compared to the earlier beta version. We divide the improvements into three categories: significant, medium, and smaller improvements.

Significant improvements:

- More data: StateParl now contains 14,092,664 cases (compared to previously 9,534,937 cases, in the earlier beta version, available at <https://doi.org/10.7802/2744>).
- Linkage with StatePol: In StateParl, persons are identified using *Länder*-specific patterns. In this version, we automatically matched these results with StatePol, the comprehensive dataset of German state politicians. Validation of the matching process (sample size: 3,330 distinct speakers) confirmed an accuracy rate of 99.34% correct matches, 0.24% unclear cases, and 0.45% false matches. Overall, 4,674 speakers in StateParl could be identified and successfully linked with StatePol.
- Introduction of a protocols dataset: The metadata of stenographic protocols is now stored in a dedicated database and provided as a separate dataset.
- Access to raw data files: StateParl now offers direct access to the originally scraped PDF files, providing full transparency and enabling further analysis.
- Correction of issues related to "Zwischenfragen" (interposed questions) in Schleswig-Holstein: The beta version of StateParl encountered speaker misidentifications due to missing speaker names after "Zwischenfragen". This issue has been addressed by storing paragraph-based position metadata, allowing for the correct identification of indents and preventing misattributions.
- Removal of Selenium-based scraping: For a number of parliaments, we have replaced Selenium scraping with more reliable data extraction endpoints, improving both efficiency and stability.

Medium improvements:

- New affiliations: "Regenbogen – Für eine neue Linke" (rgn), "Partei Rechtsstaatlicher Offensive" (pro), "Bürger für Mecklenburg-Vorpommern" (bmv)
- New affiliation aliases: "bfth" for Bürger für Thüringen [bft]; "gal" for Bündnis 90/Die Grünen [grn]; "Freie Wähler/BMV" for Freie Wähler [frw]
- Introduction of state-specific affiliation dictionaries: The processing architecture was updated to offer the flexibility of state-specific affiliation dictionaries. This allows to provide a more selective affiliation aliases in comparison to the general list. This is useful especially for regexes identifying speakers. If a state-specific dictionary is available, it will be used. Otherwise, the general dictionary will be applied.
- Remove exclamation by spaces. Names like "Renate H o l z n a g e l" can now be normalized to "Renate Holznagel"
- Improved stub cleanup: By introducing state-specific header and footer margins, we were able to remove a lot of stubs like page headers as well as name repetitions on new pages ("(Kai Vogel)").
- Removing the variables "seq_paragraphs" and "seq_speeches", and introducing the new variable "sequence_number": This version of StateParl comes with one column counting paragraphs for each protocol.

Smaller improvements:

- Fix greedy Regex catching “DIE” (Hamburg): In the previous version of StateParl, in Hamburg, “Sabine Boeddinghaus DIE” was matched when a paragraph starts with “Sabine Boeddinghaus DIE LINKE:”. This is fixed by using a state-specific affiliation dictionary which does not contain “linke” anymore.
- Improved identification of members of government (Baden-Württemberg): The patterns for identifying government members was improved for “*Staatsrätin*”, “*Staatsrat*”, “*Staatssekretär*”, “*Staatssekretärin*”, “*Ministerialdirektor*”, and “*Ministerialdirektorin*”
- Performance improvements (all *Länder*): Handling of database connections have been optimized, and queries have been improved.

5. Codebook

Paragraphs Dataset

Variable Names	Data Type ⁸	Description	Manifestations/ Examples
id	int	Unique identifier in StateParl	For example: [14574812]
protocol	varchar	Unique identifier of each stenographic protocol	For example: [be_18_84]
state	char(2)	Parliament in which plenary session took place	16 state parliaments: [bw]= Baden-Württemberg [by]= Bavaria [be]= Berlin [bb]= Brandenburg [hb]= Bremen [hh]= Hamburg [he]= Hesse [mv]= Mecklenburg-Vorpommern [ni]= Lower Saxony [nw]= North Rhine-Westphalia [rp]= Rhineland-Palatinate [sl]= Saarland [sn]= Saxony [st]= Saxony-Anhalt [sh]= Schleswig-Holstein [th]= Thuringia
period	int	Legislative period according to stenographic protocol	For example: [18]
nth	int	Plenary session within legislative period according to	For example: [82]

⁸ *int* describes a positive natural number. *date* describes a stored date element in the format YYYY-MM-DD. *char* describes a fixed-length string. *varchar* describes a string flexible in its length. *text* describes a text column variable in length.

		stenographic protocol	<i>Note: This would be the 82nd plenary session in the 18th legislative period of a parliament.</i>
date	date	Date of the plenary session according to stenographic protocol	For example: [2021-09-16] <i>The plenary session took place on 16th of September 2021.</i>
sequence_number	int	Incremental number assigned to each paragraph in stenographic protocol	For example: [22] <i>This is the 22nd paragraph in a given protocol.</i>
speaker_extracted	varchar(120)	Name of the speaker, extracted from stenographic protocol	For example: [Raed Saleh]
speaker_id	varchar(120)	Speaker identifier if identified in StatePol	For example: [be_18_spd_raed-saleh]
speaker_name	varchar(120)	Speaker name in StatePol	For example: [Raed Saleh]
affiliation	varchar(23)	Affiliation of the speaker according to stenographic protocol	<u>Parliamentary party groups:</u> [cdu]= Christlich Demokratische Union Deutschlands (CDU) [csu]= Christlich-Soziale Union in Bayern (CSU) [spd]= Sozialdemokratische Partei Deutschlands (SPD) [fdp]= Freie Demokratische Partei (FDP, F.D.P., FDP/DVP, Freie Demokraten) [grn]= Bündnis 90/Die Grünen (Grüne, Bündnisgrüne, B 90/Grüne, Grüne/B90)

			[lin]= Die Linke (Linke, Linksfraktion, Saar-Linke, LL-PDS, PDS, Linkspartei.PDS, Linksfraktion.PDS, PDS, Die Linkspartei.PDS) [afd]= Alternative für Deutschland (AfD) [frw]= Freie Wähler (FW, BVB/FW) [biw]= Bürger in Wut [bmv]= Bürger für Mecklenburg-Vorpommern [bft]= Bürger für Thüringen [büd]= Bündnis Deutschland [dep]= Deutsche Partei [dvu]= Deutsche Volksunion (DVU) [fam]= Familien-Partei Deutschlands (Familien-Partei) [fvp]= Freiheitliche Deutsche Volkspartei (FDVP) [gra]= Die Grauen – Graue Panther (Graue Panther) [lkr]= Liberal-Konservative Reformer (LKR, Allianz für Fortschritt und Aufbruch, ALFA) [npd]= Nationaldemokratische Partei Deutschlands (NPD, Heimat, Heimat!) [ödp]= Ökologisch-Demokratische Partei (ÖDP) [par]= Die Partei [pir]= Piratenpartei Deutschland (Piraten) [pro]= Partei Rechtsstaatlicher Offensive [rep]= Die Republikaner (Republikaner) [rgn]= Regenbogen – Für eine neue Linke [ssw]= Südschleswigscher Wählerverband [mrf]= M.R.F. (Gruppe Magnitz, Runge, Felgenträger)
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			[lfm]= L.F.M. (Gruppe Löhmann, Felgenträger, Magnitz) [ind]= not part of a parliamentary party group (fraktionslos) <u>Functional descriptions:</u> [pre]= presidium (Präsidium) [gov]= state government (Landesregierung) [oth]= others (such as ombudsmen and -women, data protection officers, presidents of Constitutional Courts, presidents of Court of Auditors, and other guests such as representatives of NGOs) [ijn]= all paragraphs that are not part of parliamentary speech (like interjections) are assigned to this affiliation.
content	text	Speech in the paragraph according to stenographic protocol	[Nehmen wir zum Beispiel das große Feld der Arbeit. Faire Löhne sind eine ganz zentrale Voraussetzung für ein besseres Leben. Hier sind wir in den vergangenen fünf Jahren ein Stück weitergekommen.]⁹

⁹ Translation [by the authors]: “Take the large field of labor policy, for example. Fair wages are a key prerequisite for a better life. We have made some progress in this area over the past five years.”

Protocols Dataset

Variable Names	Data Type	Description	Manifestations/ Examples
id	varchar(255)	Unique identifier of each stenographic protocol	For example [be_18_84]
state	char(2)	Parliament in which plenary session took place	For example [be]
period	int(11)	Legislative period according to stenographic protocol	For example: [18]
nth	int(11)	Plenary session within legislative period according to stenographic protocol	For example [84]
date	date	Date of the plenary session according to stenographic protocol	For example [2021-09-16]
url	varchar(255)	Direct URL to the PDF file on the web service ¹⁰	For example [https://pardok.parlament-berlin.de/starweb/adis/citat/VT/18/PlenarPr/p18-084-wp.pdf]

Mandate Mappings Dataset

Variable Names	Data Type	Description	Manifestations/ Examples
stateparl_mandate_id	varchar(255)	Unique identifier per mandate	For example: [be_18_spd_raed-saleh]
statepol_mandate_id	varchar(255)	StatePol mandate identifier	For example: [/wiki/Raed_Saleh-be-18]

¹⁰ These URLs are subject to frequent changes. This is why we also provide access to the PDF files of the stenographic protocols.